

1 Introduction

Global Navigation Satellite Systems (GNSS) receivers are essential for determining the position, velocity, and time of an object. However, there are several sources of error that can affect the accuracy and reliability of GNSS signals, causing inaccuracies in their ability to provide accurate positioning. And atmospheric errors are important to consider.

Atmospheric errors occur when signals from GNSS satellites pass through the Earth's atmosphere. This can cause the signals to be delayed or refracted, resulting in errors in the receiver's position. This means that atmospheric corrections are critical to GNSS observations because they significantly improve the accuracy and reliability of positioning data.

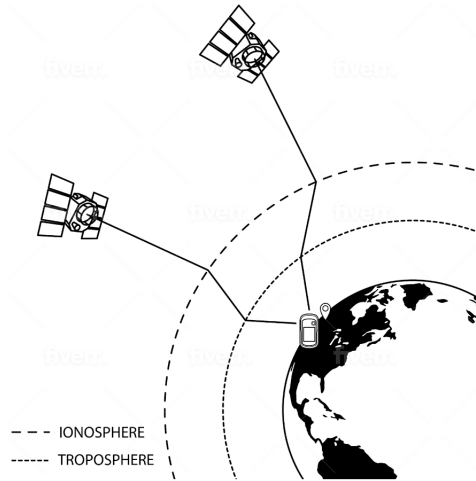


Figure 1: Atmospheric errors.
Source: Global GPS Systems [4].

The troposphere is the layer of the atmosphere between the Earth's surface and an altitude of about 60 kilometers.

The effect of the troposphere on GNSS signals appears as an additional delay in the measurement of the signal as it travels from the satellite to the receiver. This delay depends on temperature, pressure, humidity, and the location of the transmitter and receiver antennas.

The main characteristic of the troposphere is that it is a non-dispersive medium for electromagnetic waves up to 15 GHz, i.e. the tropospheric effects are not frequency dependent for GNSS signals. Therefore, the carrier phase and code measurements are affected by the same delay.

A direct consequence of the non-frequency-dependent delay is that tropospheric refraction cannot be eliminated by combining dual-frequency measurements. Therefore, the only way to mitigate the tropospheric effect is to use models and/or estimate it from observations.

The tropospheric delay depends on the signal path through the neutral atmosphere and can then be modeled as a function of satellite elevation angle. This elevation angle dependence means that we must assign a value to the associated delay, the Zenith Tropospheric Delay (ZTD), which is the delay experienced by GNSS signals traveling directly overhead (at the zenith) due to the troposphere. This delay is caused by the refraction of signals as they pass through the troposphere, which contains both dry gases and water vapor. And it can lead to a positioning error of a few meters. Because

of the differences between the atmospheric profiles of dry gases and water vapor, it is better to use different mappings for the dry and wet components.

One of the most common methods used to compute the tropospheric delay, is given by the Saastamoinen tropospheric model, this model divides the total tropospheric delay into two main components: hydrostatic delay and wet delay. The hydrostatic delay is primarily caused by the dry gases in the atmosphere, while the wet delay is due to the water vapor in the atmosphere. This last one, is more variable and depends on the humidity and temperature. The total delay is the sum of the hydrostatic and wet delays.

[2]

The ionosphere is the zone of the terrestrial atmosphere that extends from about 60 km to over 2,000 km high. As its name implies, it contains a partially ionized medium as a result of X- and UV-radiation from the Sun and the incidence of charged particles.

The propagation speed of GNSS electromagnetic signals in the ionosphere depends on its electron density, which is typically driven by two main processes: during the day, solar radiation causes the ionization of neutral atoms, producing free electrons and ions. During the night, the recombination process dominates, where free electrons recombine with ions to produce neutral particles, leading to a decrease in electron density.

Ionospheric delay occurs because the ionosphere is a dispersive medium, meaning that the speed at which GNSS signals travel through the ionosphere depends on the signal frequency. This delay can lead to errors in the calculated positions of GNSS receivers. The delay is more pronounced during the day when solar radiation is stronger and ionization levels are higher. Phase measurements suffer from a negative delay due to the advance through the ionosphere, and code measurements suffer from a positive delay.

Because the ionosphere is a dispersive medium, the refraction of GNSS signals depends on their frequencies (as the inverse square). This dependency on signal frequency allows us to remove most of its effect by using dual-frequency measurements. However, single-frequency receivers must apply an ionospheric prediction model to remove (as much as possible) this effect, which can reach tens of meters. In addition, ionospheric refraction depends on the geographic location of the receiver, the time of day, and solar activity. [1]

Understanding the methods to perform this atmospheric correction is a key element in achieving accuracy and reliability of GNSS signals, which is very important for applications requiring accurate positioning. The goal of this exercise is to understand how the troposphere and ionosphere affect GNSS observations and to develop appropriate methods that can be used to correct these errors in the signals received by a GNSS station with known coordinates, thereby achieving higher accuracy and reliability in GNSS systems. At the same time, we will study how atmospheric parameters such as temperature and humidity affect the tropospheric delay.

2 Methodology

To accomplish the goals of the previous assignment, the expressions and methods necessary to correct the atmospheric errors should be presented. The explanations will be divided into two sections, one for each focus of the atmospheric errors.

Tropospheric delay

To compute the tropospheric delay, we will use the Saastamoinen tropospheric model, which takes into account atmospheric parameters such as pressure, temperature, and humidity, and the zenith value of the satellites to compute this error.

To obtain the zenith angle for each of the satellites, we will use the ENU coordinates of each satellite computed using the Cartesian coordinates of the station as the origin of the coordinates.

$$\text{Zenith angle, } z = \cos^{-1} \left(\frac{u}{\sqrt{e^2 + n^2 + u^2}} \right)$$

For the atmospheric values, we need to know the temperature in Kelvin. Any temperature in Celsius degrees can be converted to Kelvin by adding 273.15 to the degree values. The atmospheric pressure must be given in millibars. From the value of the relative humidity, RH, we can calculate the partial pressure of water vapor, e_s , in hectopascals, as

$$e_s = 6.108 \cdot RH \cdot \exp \left(\frac{17.15T - 4684}{T - 38.45} \right).$$

We convert this value to millibars considering that 1 hPa = 1 mbar.

To get the tropospheric delay in meters, it is modeled according to the Saastamoinen model as

$$\Delta^{Trop} = \frac{0.002277}{\cos z} \left[p + \left(\frac{1255}{T} + 0.05 \right) e - \tan^2(z) \right].$$

However, this method still needs some refinement, which could be achieved by introducing two correction terms B and δR . The first term takes into account the height of the observation site, this term accounts for the variation in atmospheric pressure and water vapor content at different heights. The second term takes into account altitude and zenith angle, which corrects for additional errors due to variations in atmospheric refraction with altitude and viewing geometry.

The computation of both B and δR is achieved by interpolation and extrapolation using the values of these parameters given in [3]. Through the interpolation process, we take the closest value given the station altitude and zenith angle.

The final expression for the Saastamoinen model will be

$$\Delta^{Trop} = \frac{0.002277}{\cos z} \left[p + \left(\frac{1255}{T} + 0.05 \right) e - B \tan^2(z) \right] + \delta R$$

In Matlab, we created two different Matlab functions, **troposphericdelayuncorrected**, which calculates the tropospheric delay from the zenith angle of the satellites and the atmospheric parameters, without the correction parameters. The second function, **troposphericdelay**, calculates the tropospheric delay with the correction parameters from the zenith angle of the satellites, the atmospheric

parameters, and the station altitude. This function uses interpolation to compute the values of the correction parameters.

To check the correctness of the obtained values of the tropospheric delay, we have check with our functions the results obtained whith the parameters of standard atmospheric levels at sea level, and we have verify that the tropospheric delay is around 2.3 for zenith angle equal 0° .

In addition, to study the influence of temperature and humidity on the tropospheric delay, we have created a range of values for these parameters and we have computed the corresponding water vapor partial pressure and the value obtained by the Saastamoinen model for these different values of temperature and humidity. For each satellite, we have plotted the variation of the Saastamoinen model against the change in temperature and humidity.

Ionospheric delay

Considering the L1 frequency commonly used in GNSS systems such as GPS, we will introduce the corresponding expressions for the ionospheric delay.

$$I_f^\zeta = I_f^{Zenith} \cdot OF_\xi$$

This expression takes into account the first-order ionospheric delay I_f^{Zenith} , which is the dominant component of the ionospheric effect on GNSS signals, causing the signal to be delayed or advanced depending on the frequency. And also the obliquity factor OF_ξ , which accounts for the angle the GNSS signals follow as they travel through the ionosphere.

$$I_f^{Zenith} = \Delta_{gr}^{iono,zenith} = \frac{40.3}{f^2} TEC$$

$$OF_\xi = \left(1 - \left(\frac{R_E \sin \zeta}{R_E + h_I} \right)^2 \right)^{-1/2}$$

For the first-order ionospheric delay I_f^{Zenith} we have that, f is the carrier frequency, and for the L1 frequency it takes a value of $f_{L1} = 1575420000\text{Hz}$, and TEC is the Total Electron Content, which measures the number of free electrons along the signal path, and it takes values of the order of $1e16$. To get the exact value of the TEC, we search the register values in [5] for the coordinates of the station and for the time of the observation 2004-02-02 01:00:00, and we get that the TEC value is $2.5 \cdot 1e16$. The image 2, is a graph of the TEC values.

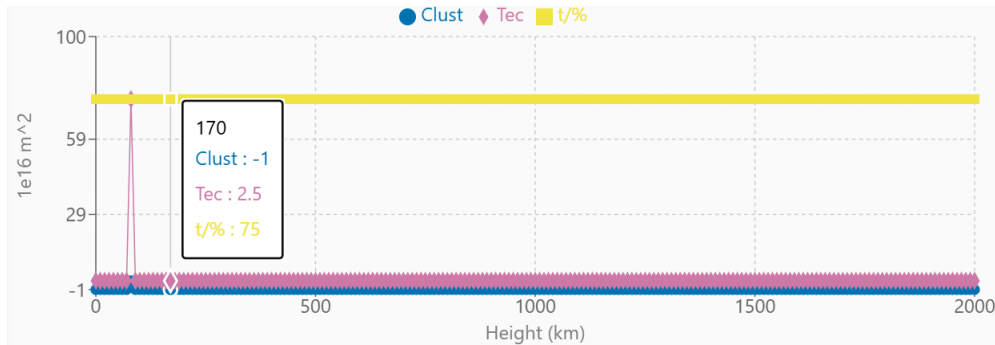


Figure 2: TEC values.

Source: *International Reference Ionosphere (IRI)* [5].

For the obliquity factor OF_ζ we have that R_E is the mean radius of the Earth, which takes a value of 6371000 meters; $h_I = 350000\text{m}$ is the mean ionospheric height, and the zenith angle is ζ .

To compute the values of the ionospheric delay, we have created a Matlab function `ionosphericdelay` that, given the values of frequency, TEC, mean ionospheric height, Earth mean radius, and zenith angle, returns the ionospheric delay in meters.

3 Results

In this section we will introduce the results obtained for each of the atmospheric errors.

For both tasks, it is necessary the computation of the zenith angle, to do it, we will obtain the ENU coordinates of the satellites considering the center of coordinates at the station whose Cartesian coordinates are [3104219.4530, 998383.9820, 5463290.5080].

Tropospheric delay

We have compute for each satellite the tropospheric delay and the refined tropospheric delay corrected by the parameters B and δR . In table 1 we can see the comparison between the both calculations and we observed that the values corrected tropospheric delay are slightly higher due to the addition of the term δR . At the same time, we observe that higher zenith values, cause a higher delay, this is because the tropospheric delay depends on the length of the signal path through the atmosphere, and, a higher zenith angles is due to a longer slant path through the atmosphere.

Satellite number	Tropospheric delay [m]	Corrected Tropospheric delay [m]
2	9.2465	9.2580
3	10.8384	10.8558
8	2.6569	2.6579
10	2.8106	2.8115
13	5.0283	5.0287
17	5.5942	5.5965
21	13.1525	13.2124
24	14.7667	14.7409
26	12.9248	12.9781
27	2.6643	2.6652
28	12.9483	13.0013
29	5.1761	5.1763

Table 1: Comparison of Tropospheric delay (uncorrected) and Tropospheric delay with correction parameters for each satellite.

To study the influence of humidity and temperature in the tropospheric delay, we have create a range of values of temperature that goes from -30 to 30 degrees, and a variation of temperature that goes from 0% to 100%. And then we have plot the variation of the tropospheric delay according to the Saastamoinen model and the different values of temperature and humidity.

The plots in figure 3 represent the variation with the temperature of the tropospheric delay, and the plots in figure 4 represent the variation with the humidity of the tropospheric delay.

As we see, the increase given by the temperature is exponential, it starts with a smaller increase while the temperature are negative, but then we are able to see a rapid increase when the temperatures are positive in Celsius, or higher than 273.15 in Kelvin. However, the increase of the tropospheric delay given by the humidity is more constant. In both cases we see that that the minimum and maximum value of tropospheric delay are the same in this intervals of temperature and humidity. We also appreciate that there is more change in the values with higher zenith angles.

In the plots in figure 5, we have plot the change in the tropospheric delay considering both variations in temperature and humidity, and we can see that the change given by humidity starts to be significant earlier because of the more direct increase.

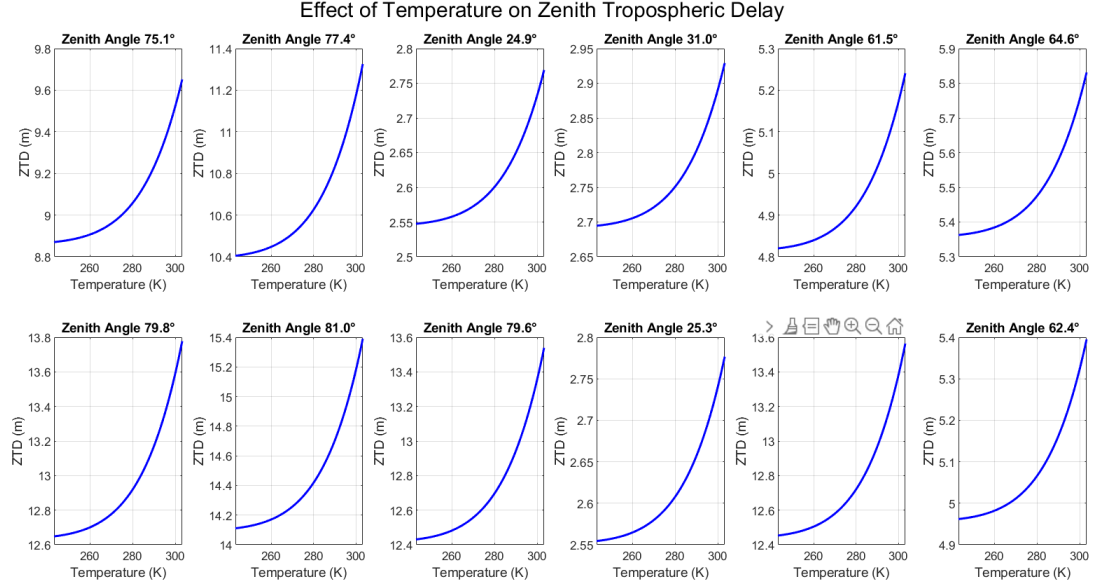


Figure 3: Change of tropospheric delay with temperature.

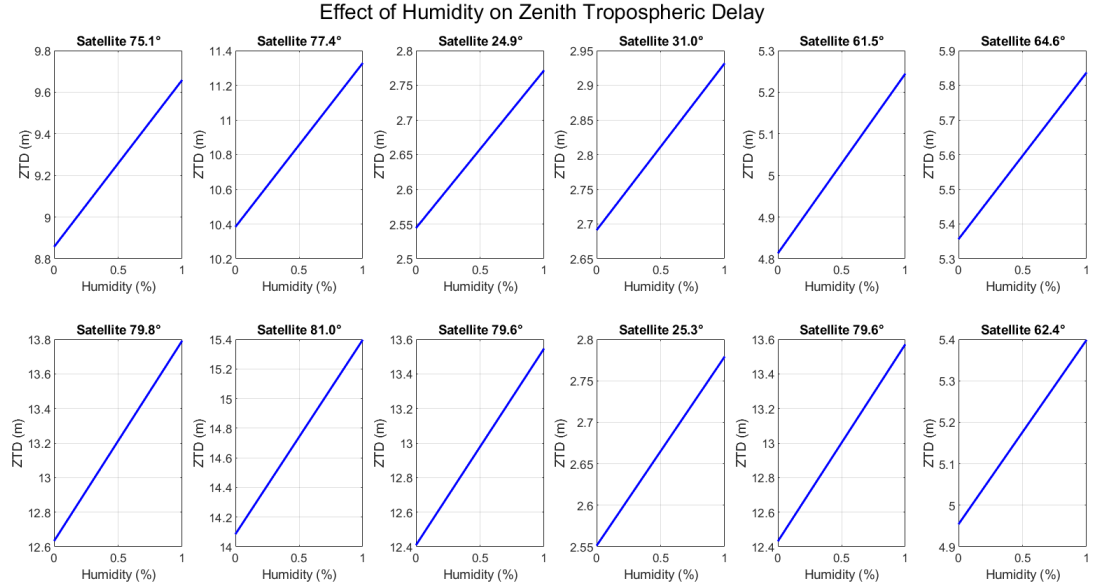


Figure 4: Change of tropospheric delay with humidity.

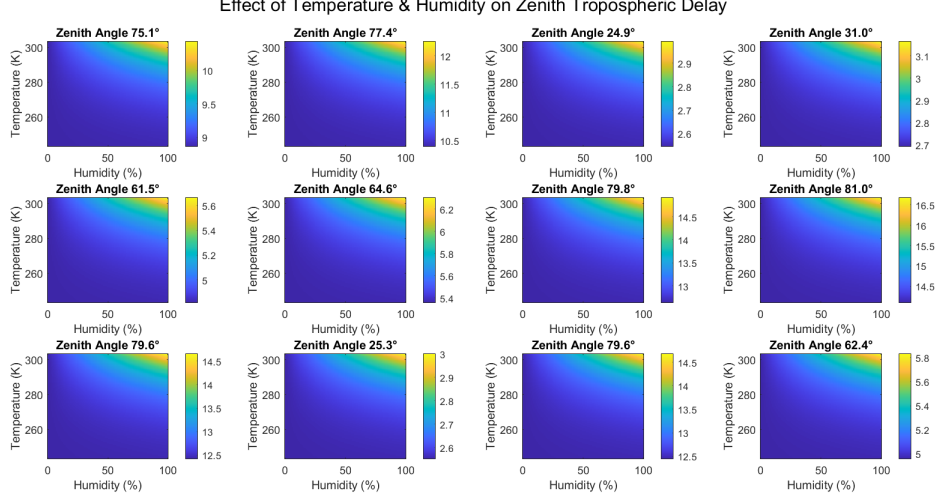


Figure 5: Change of tropospheric delay with humidity and temperature.

Ionospheric delay

Given the values of the parameters and the computed zenith angle values, we have computed the ionospheric delay in meters. The results are included in table 2. As we observe, the ionospheric delay increases as the zenith angle increases, due to the longer path the signal travels through the ionosphere.

Satellite number	Ionospheric delay [m]
2	1.012152032135862
3	1.069014854238742
8	0.442844406618627
10	0.465181370078461
13	0.733240936943523
17	0.785932255910479
21	1.126360699305451
24	1.154569675524759
26	1.121719023916396
27	0.443920494152294
28	1.122206703695352
29	0.747580504650249

Table 2: Ionospheric delay for different satellites

4 Analysis and Discussion

The signals transmitted by the GNSS satellites to the receivers are modified by the characteristics of the different layers of the atmosphere, the layers such as the troposphere or the ionosphere produce a delay in the signals that must be taken into account when processing them in order to achieve accuracy and precision in the positioning, since both delays could be translated in an error in the positioning of meters.

There are different models that can be applied to calculate these delays in order to correct the positioning. We have observed that both the ionospheric and the tropospheric delay depend on the zenith angle of the satellites, since the higher the zenith angle, the greater the distance the signal has to travel through the atmosphere. The higher the angle, the longer the path through the ionosphere, leading to greater delay.

At the same time, we have observed that the tropospheric delay is highly dependent on the atmospheric values, concluding that an increase in temperature and humidity will result in an increase in the delay. This is because, for example, humidity increases the partial pressure of water vapor (e_s), which appears as a positive multiplier in models such as Saastamoinen's and significantly affects the delay. In the case of temperature, it is mathematically explained by the exponential growth of the water vapor partial pressure with increasing temperature. Also, as the temperature increases, the amount of water vapor in the air may also increase, leading to a larger wet component of the tropospheric delay [2], this is well captured in models like Saastamoinen's.

At the same time, we have observed that the change seems to be greater in cases where the zenith angle is larger, also due to the distance the signal travels through the atmosphere, encountering more atmospheric particles and water vapor.

By understanding and correcting these atmospheric effects, that are sources of atmospheric errors that lead to positional inaccuracies, GNSS systems can achieve higher precision and reliability. Addressing these atmospheric errors is essential for applications requiring accurate positioning, from everyday navigation to high-precision scientific research.

References

- [1] European Space Agency (ESA). *Ionospheric Delay*. Accessed: March 31, 2025. 2025. URL: https://gssc.esa.int/navipedia/index.php/Ionospheric_Delay.
- [2] European Space Agency (ESA). *Tropospheric Delay*. Accessed: March 31, 2025. 2025. URL: https://gssc.esa.int/navipedia/index.php/Tropospheric_Delay.
- [3] Bernhard Hofmann-Wellenhof, Herbert Lichtenegger, and Elmar Wasle. *GNSS – Global Navigation Satellite Systems*. Vienna: Springer Verlag, 2008. ISBN: 978-3-540-72729-3.
- [4] Global GPS Systems. *GNSS Timing and Atmospheric interferences: How GNSS is Solving These Problems*. Accessed: March 31, 2025. 2025. URL: <https://globalgpssystems.com/gnss/gnss-timing-and-atmospheric-interferences-how-gnss-is-solving-these-problems/>.
- [5] International Reference Ionosphere (IRI) Team. *IRI-2020 Model*. Accessed: March 31, 2025. 2020. URL: <https://ccmc.gsfc.nasa.gov/models/IRI~2020/>.